

Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

4-quadrant silicon PIN photodiode

The Optoelectronics group of Vishay Intertechnology has introduced a 4-quadrant silicon PIN photodiode, which is claimed to be the industry's first. Available in a standard surface-mount package, it is qualified for auto-



motive applications. Combining high photosensitivity with low 0.1% crosstalk, and virtually no

tolerance between its segments, the K857PE enables a variety of sensor and control applications for the automotive, consumer, and industrial markets. Built on epitaxial technology, the sides of the K857PE eliminates stray light irradiating the photodiodes, resulting in an excellent signal-to-noise ratio. The device's linear photo response enables small signal detection for automotive rain/light sensors, industrial automation systems, laser beam alignment, and virtual reality applications. It also features a daylight blocking filter, $\pm 60^\circ$ angle of half sensitivity, and operates over a temperature range of -40°C to $+110^\circ\text{C}$.

www.vishay.com

Automotive Wi-Fi 6E module

LG Innotek has developed an automotive Wi-Fi 6E module with next-generation Wi-Fi technology. Automotive Wi-Fi 6E modules are near-field wireless communication components connecting in-vehicle infotainment (IVI) systems,



which control driving information and multimedia content for internal smart devices and external routers. With the next-generation Wi-Fi 6E (6th Generation Extended) technology using 6GHz bandwidth, the module boasts data transmission speeds approximately three times faster than the existing Wi-Fi 5. It combines a communications chipset, a radio frequency (RF) circuit, and other components. The module is mounted on multiple components of a vehicle including the audio-video navigation (AVN) system and smart antenna. It allows easier software updates of the automotive navigation system.

www.lginnotek.com

Powerful electric vacuum gripper

Complex packaging of bulky cardboard boxes are demanding while end-of-line operations such as palletising are labour-intensive and costly. Now OnRobot's new electric vacuum gripper, the VGP20,



can address that. Compatible with all leading robot brands, the gripper can handle payloads of 20kg, making it a great fit for a wide range of applications in industries from cosmetics and electronics to pharmaceuticals and food and beverages. It enables companies to save up to 90% on operating and maintenance costs compared to traditional pneumatic gripper deployments. Additionally, the VGP20 gripper's built-in intelligence, combined with its easy-to-use software, provides precise air flow control functionality, which allows users to vary the type of grip used in different applications.

<https://onrobot.com>

Ultrasmall, high-capacity MLCC

Samsung Electro-Mechanics has developed a 0402-size (length 0.4mm, width 0.2mm) 1.0 μF capacity, 6.3V internal voltage product. MLCC is a core component within electronic devices



that controls the stable flow electric current within the electronic product circuit. With the implementation of various 5G functions, including technology advancement and multi-camera, ultrasmall MLCC that can withstand high capacity and voltage is essential. Moreover, the product enhances stability of its product life and operation in terms of its DC BIAS characteristics (a characteristic that reduces product volume when DC voltage is applied; superior as there are less changes). The company maintained 1.0 μF high-capacity characteristics for 0402 ultrasmall while increasing the rated voltage by 1.5 times ($4 \rightarrow 6.3\text{V}$), allowing it to be applicable to various high-performance IT devices.

www.samsungsem.com

Small, wide-angle lidar sensor

Cepton Technologies has announced its path-breaking miniature lidar sensor, the Nova, for high-volume automotive applications. The Nova is a wide-angle, near-range lidar sensor intended to address major gaps in proximity detection of objects with current sensor technologies. It features best-in-class, high-resolution 3D imaging with a high field of view of $90\text{--}120^\circ$ (H) and $60\text{--}90^\circ$ (V), depending on configuration. The non-rotational, mirrorless, and frictionless lidar can be hidden all around a